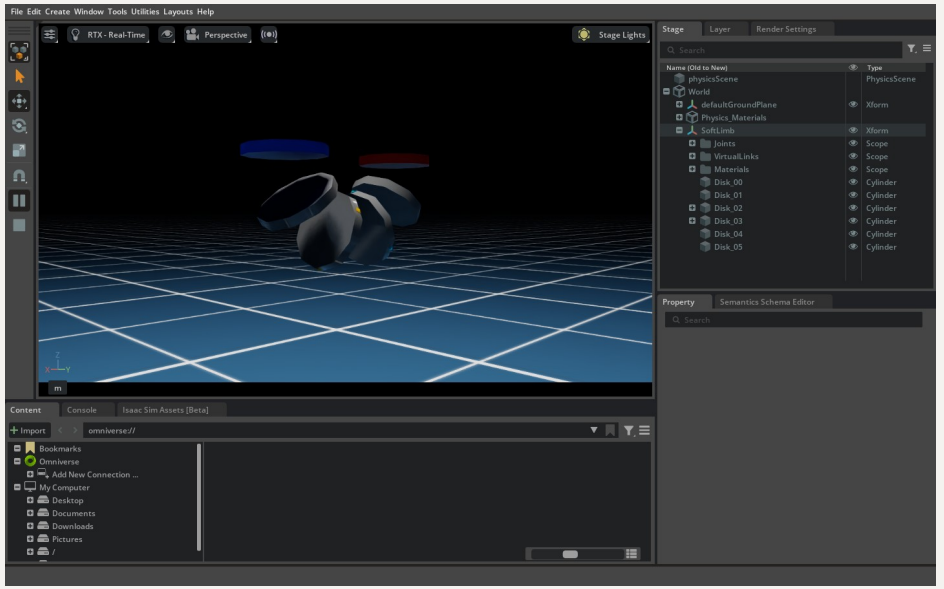
Soft Continuum Limb: Research Progress Update

Progress after successful single-module position control

Six-disc constant-curvature simulation • hybrid torque control • virtual strain sensing • Wheatstone-bridge output • RViz reconstruction



Current verified endpoint

1.0 N·m → 179.9999°

September 2026

Starting Point and Scope of This Update

✓
Single-module position control established

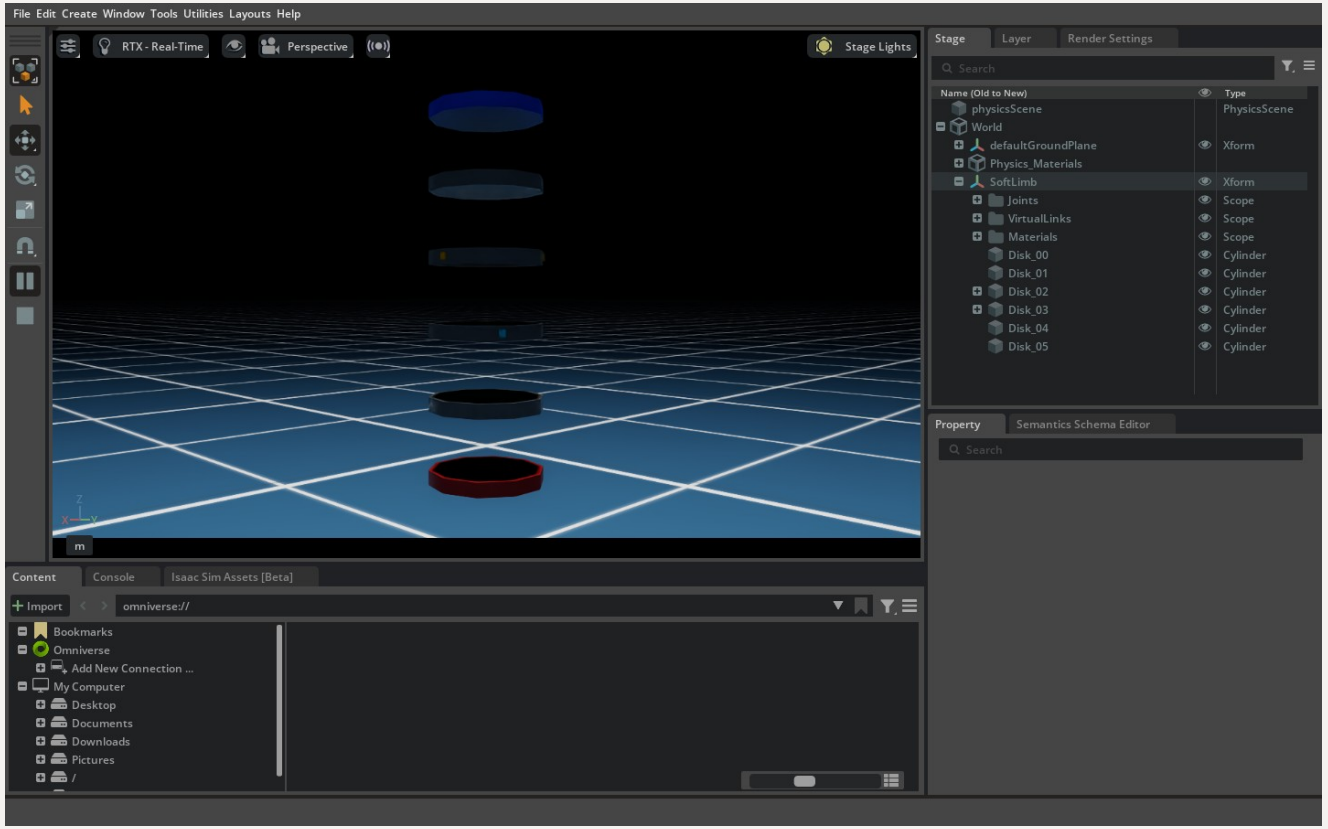
6 discs
Current reduced-order limb architecture

180°
Active maximum bend in both interfaces

Work completed after the baseline milestone

- Converted the twelve-disc normalization logic to a six-disc, five-interval model.
- Synchronized all physical intervals so the limb assumes one curve during motion.
- Added torque-commanded admittance while preserving position-control operation.
- Added two rings of three pose-derived virtual strain gauges and bridge voltages.
- Published robot_description and joint states for sensor-driven RViz visualization.
- Added reproducible regression tests and a continuously updated research record.

Current Six-Disc Working Architecture

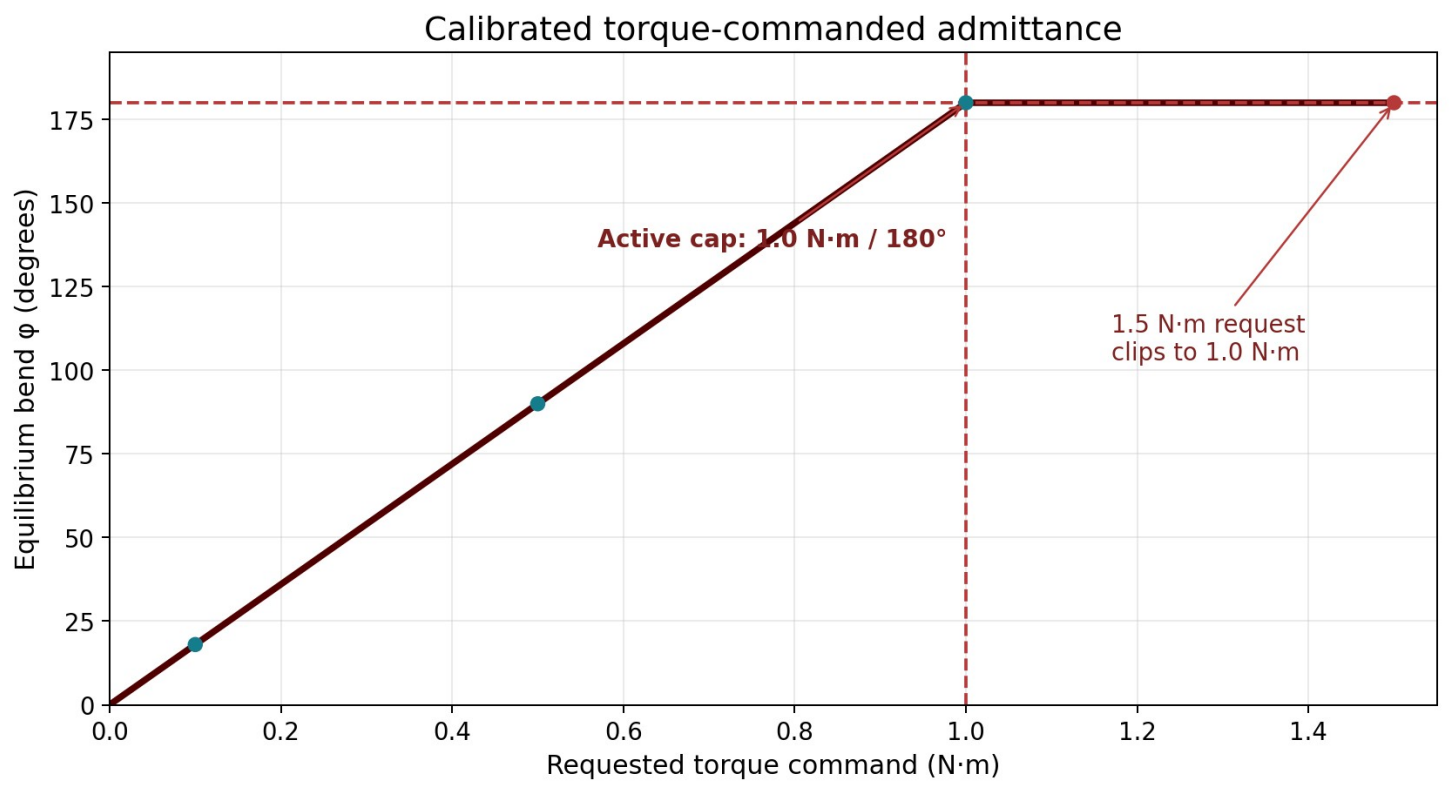


Working configuration

- Six physical discs: Disk_00 through Disk_05
- Five physical bending intervals
- Hanging base with active gravity compensation
- Direct runtime D6 joint per interval
- 240 Hz physics and 60 Hz rendering
- Overdamped position drives: stiffness 6400, damping 240
- Twist locked; bend direction controlled by θ

Captured from the current Isaac Sim working launch.

From Position Control to Hybrid Torque Control



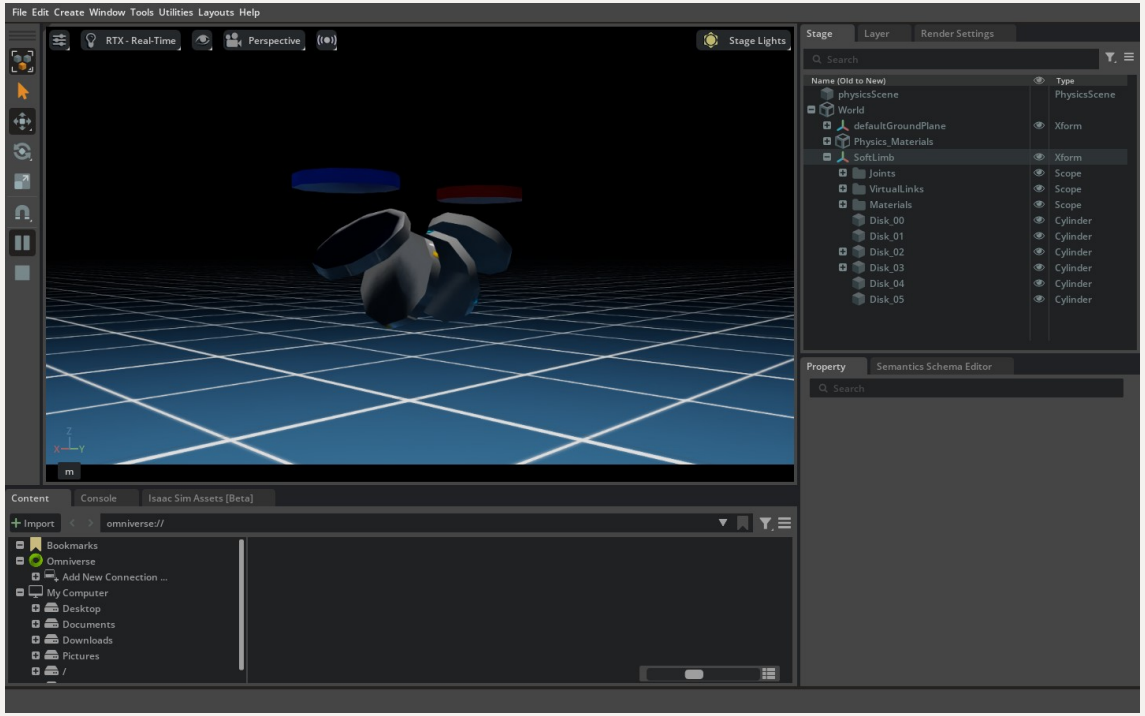
Implemented control law

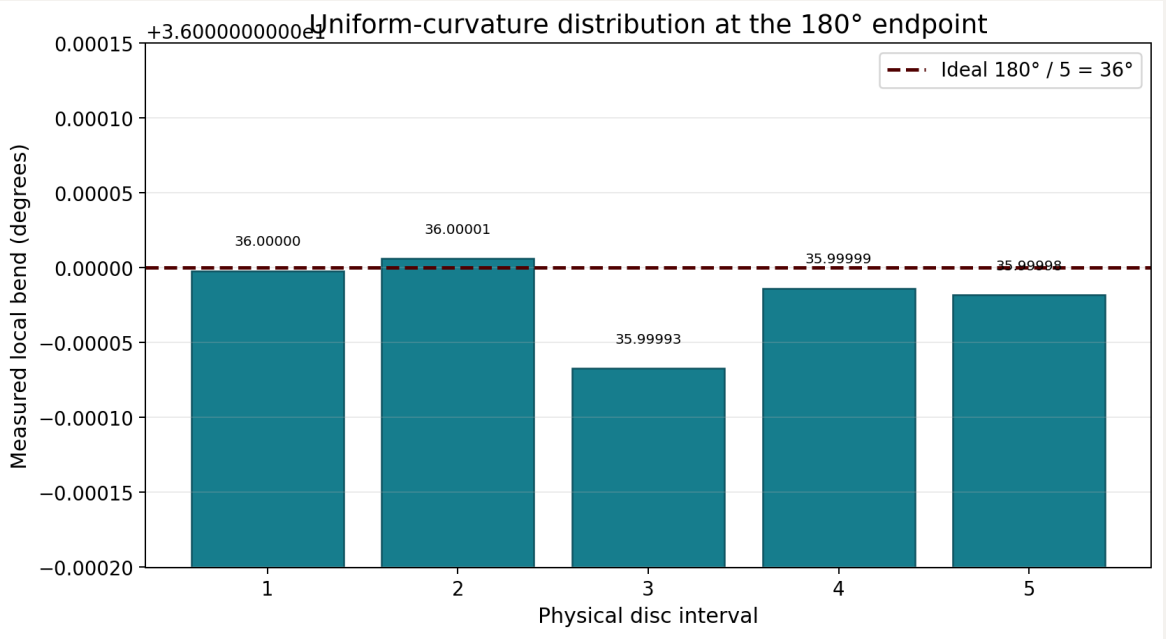
$$\varphi_{eq} = 180° \cdot \tau_c / 1.0\ \text{N·m}$$

- 0.1 N·m → 18°
- 0.5 N·m → 90°
- 1.0 N·m → 180°
- Commands beyond ±1.0 N·m are clipped
- 2.0 N·m/s command slew limit
- Position commands are ignored while nonzero torque is active

This is calibrated torque-commanded admittance—not raw open-loop torque and not a physical moment–curvature calibration.

Constant-Curvature Motion Is Enforced Across the Limb



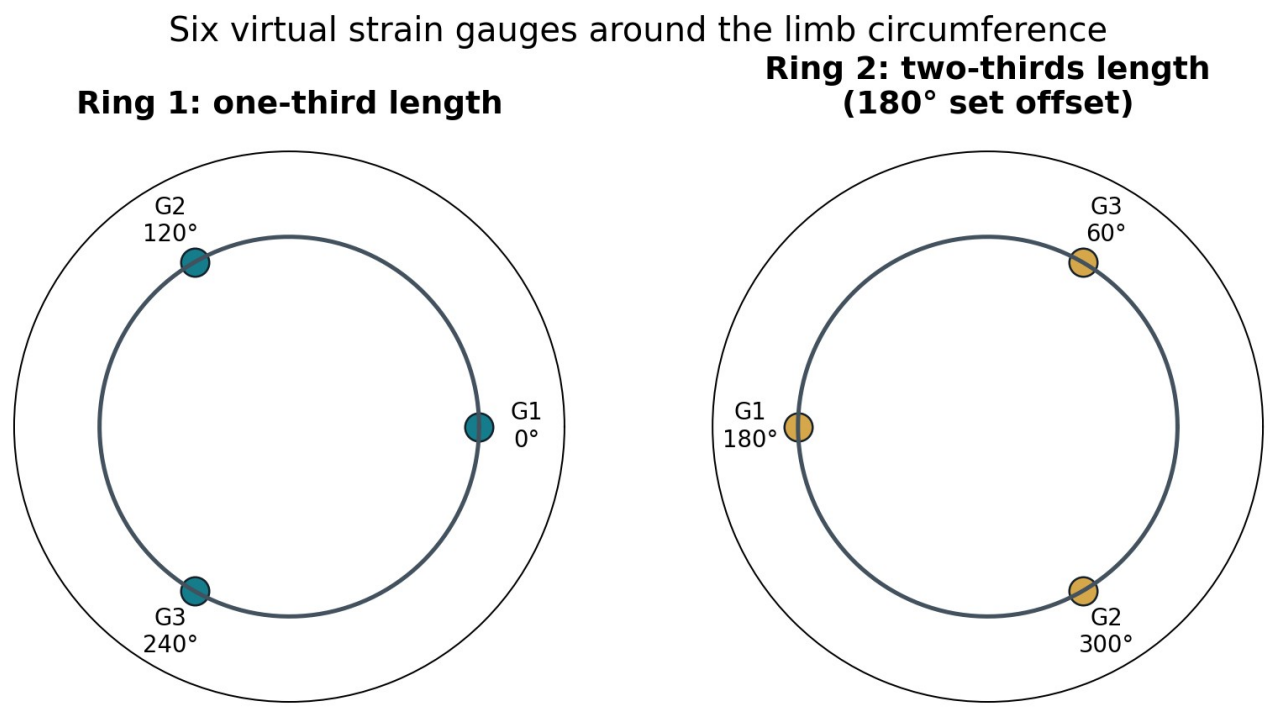


≈36° each
Five local rotations at the 180° endpoint

0.0001°
Measured torque-mapping tracking error

The controller distributes one total bend uniformly; synchronized target slew prevents the former segment-by-segment transient.

Six Virtual Strain Gauges



Placement

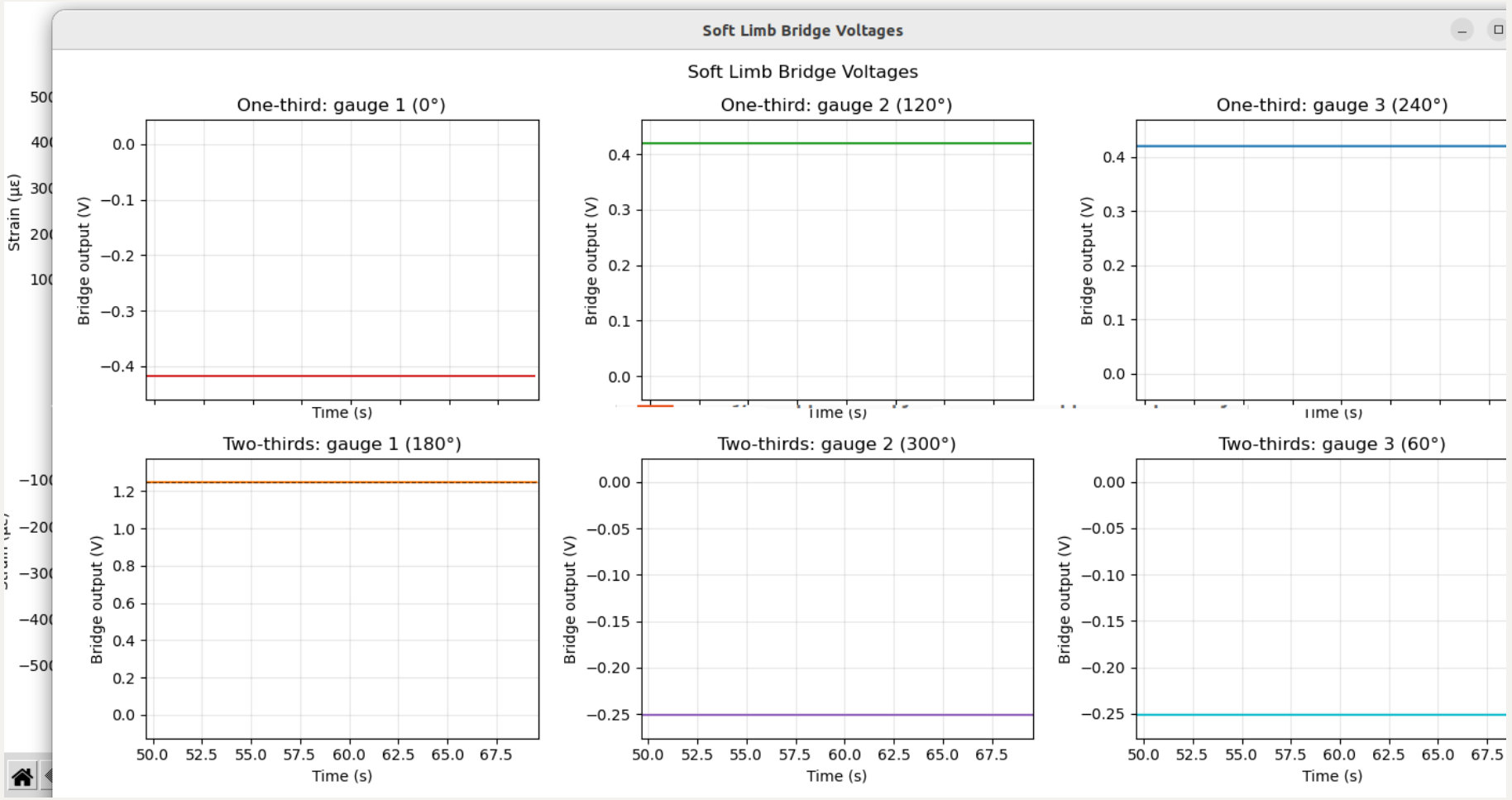
- Ring 1 at one-third length: 0°, 120°, 240°
- Ring 2 at two-thirds length: 180°, 300°, 60°
- Second ring uses the requested 180° set offset
- Readings derive from live disc poses—even during manual dragging

Implemented surface-strain model

$$\varepsilon(\alpha) = r\,[-\kappa_x \sin(\alpha) + \kappa_y \cos(\alpha)]$$

The readings are kinematic, pose-derived estimates. Physical gauge bonding, material hysteresis, temperature drift, and calibration remain future work.

Live Six-Channel Strain Response

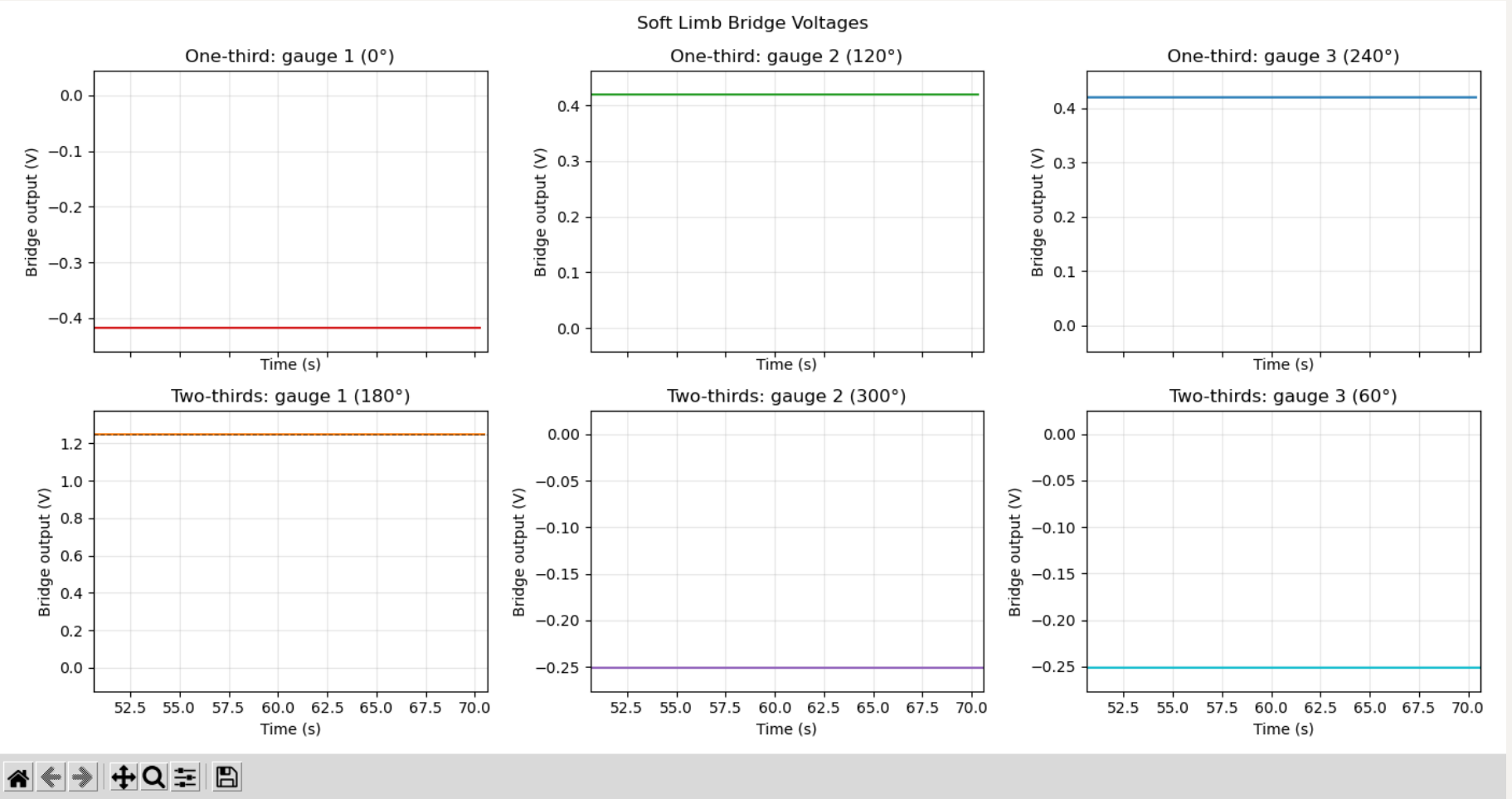


Acquisition

- Captured from live ROS topics
- Torque stepped through 0, 0.5, and 1.0 N·m
- Two stations × three circumferential gauges
- Opposing signs distinguish tension and compression
- Signals remain available without a control command

Simulated, pose-derived strain; not yet hardware validated.

Wheatstone-Bridge Voltage Layer



Nominal bridge model

- 350 Ω foil gauge
- Gauge factor GF = 2.0
- Quarter-bridge topology
- Excitation Vexc = 2.5 V
- Validity flags on separate ROS topics
- Non-finite model regions displayed at a marked 1.25 V boundary

The voltage layer supports DAQ planning; it does not establish a purchased sensor's survivability or experimental accuracy.

ROS 2 and RViz Visualization Pipeline



- robot_description is published at simulator launch.
- Joint states reconstruct the sensed limb rather than displaying a second physical Isaac limb.
- A static map → soft_limb_disk_00 transform supplies the RViz fixed frame.

Verification Summary

179.9999°
1.0 N·m endpoint

PASS
1.5 N·m request clipped to 1.0 N·m

≈36°
Every physical interval

0.0001°
Torque mapping error

Additional verified behavior

- Position control still reaches the 180° endpoint.
- First-step X/Y targets are identical across all five physical intervals.
- The sixth disc is treated as the final disc—not as a midpoint from the old twelve-disc model.
- Syntax, topology, bridge-model, ROS build, Isaac regression, and LaTeX builds pass.

Research interpretation

These results validate the implemented software mapping in unloaded, single-axis simulation. They do not yet validate material behavior, payload response, physical torque requirements, or real strain gauges.

What Changed in the Latest Iteration

Previous development state

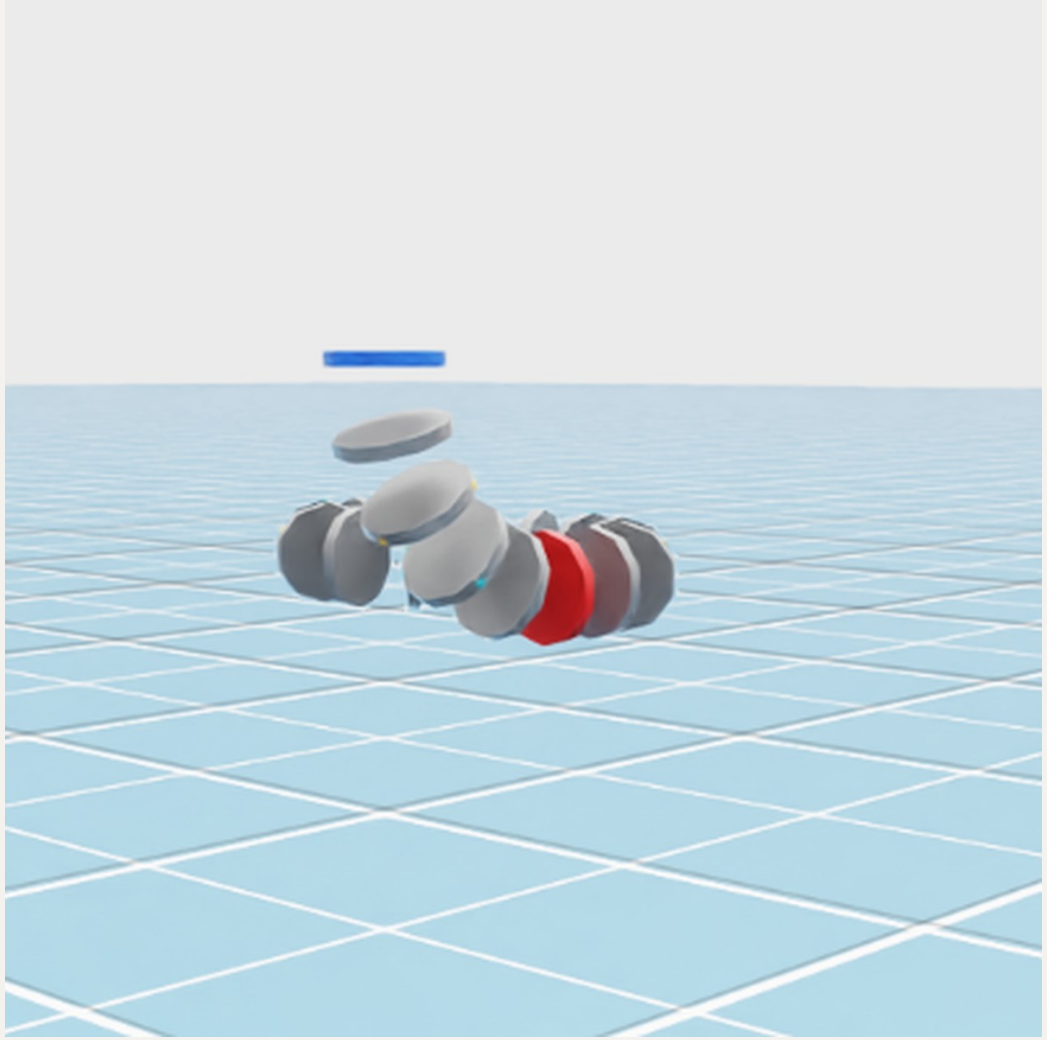
- Torque range temporarily extended to 1.5 N·m
- Accumulated bend measurement demonstrated 270°
- Useful as historical topology evidence
- Outside the intended active research range

Current active working state

- Maximum bend restored to ±180°
- Torque command capped at ±1.0 N·m
- Over-limit commands clamp safely and visibly
- Verifier checks requested vs. applied torque
- README, validation log, and research document synchronized

Decision: retain the 270° result only as historical pre-cap evidence; use 180° as the active experimental boundary.

Third Objective: Three Modules in Series



soft_3_working architecture

- Three serial six-disc modules: 18 physical discs
- Five actuated intervals per module: 15 total
- Two fixed connectors transmit pose between modules
- The first disc is fixed in air; the chain hangs under gravity
- Three independent φ-torque commands
- Three independent θ bending directions
- Same synchronized constant-curvature target inside every module
- Source, launch, control, plotting, verification, and USD files use the soft_3_working name

Live Isaac Sim capture using the Grey Studio light rig; overlapping discs reflect the commanded compound three-dimensional pose.

Independent Three-Module Control: Verified

90.0047°
Module 1: 0.50 N·m, θ = 0°

134.9999°
Module 2: 0.75 N·m, θ = 90°

179.9999°
Module 3: 1.00 N·m, θ = −90°

12-second headless regression

- All five first-step interval targets matched within each module.
- Final-second ranges: 0.0001642°, 0.0000568°, 0.0000713°.
- Connector residual bends: 0.0000672° and 0.0000394°.
- Detected all 18 gauge publishers.
- Overall verifier result: PASS.

Research boundary

This validates independent software commands, synchronized interval targets, connector rigidity, and numerical stability. It does not calibrate physical torque, material response, or payload capacity.

Minimum-Energy Horizontal Tip Circle

212.34 mm
measured mean radius

0.71 mm
height peak-to-peak

0.1254 N·m
maximum module command

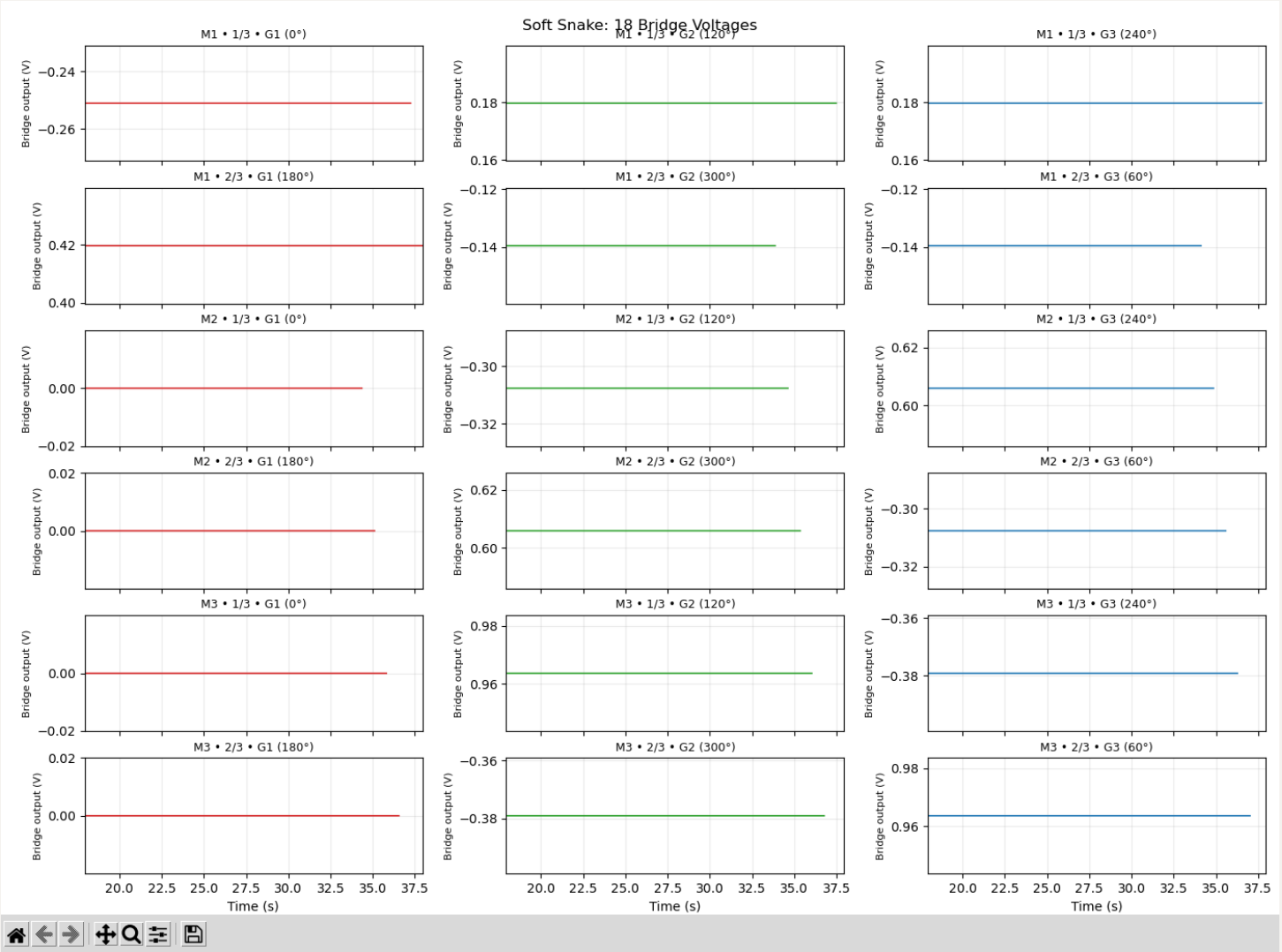
Constrained constant-curvature solution

- Minimizes $0.5\ \Sigma\ w_i\tau_i^2$ subject to requested tip radius and height.
- All 15 bending intervals and both rigid connector spans are modeled.
- Optimized torques stay constant while the common bending plane rotates.
- Default radius: 212.5 mm = half of the 425 mm modeled limb length.
- Path height is 0.40 m; the 54 s period preserves the prior tip speed.
- Full revolution passed: 0.76 mm radial and 0.71 mm height variation.

Research boundary

Energy is a normalized squared-command/elastic proxy. It is not electrical consumption, actuator work, or calibrated mechanical energy in joules.

Eighteen-Channel Bridge-Voltage Instrumentation



Repeated sensing layout

- Six gauges per module
- Two axial stations
- Three gauges per ring
- 18 bridge-voltage channels
- 18 validity flags
- Quarter bridge: 350 Ω, GF 2.0, 2.5 V excitation
- All 18 voltage graphs open automatically with the simulation launch
- Pose-derived readings remain available without a command

Live voltage graph captured from /soft_snake/module_N/sensors/... topics.

Twelve-Disc Quadruped: Hybrid Control

5
12-disc robot members

11
physical intervals per member

1 N·m → 180°
command-admittance scale

Stable original assembly with a new control interface

- Working simulator regenerated directly from the old grounded quadruped.
- USD, height, positions, ±90° placement, attachments, and camera are retained.
- Left legs: net 180° axial rotation; right legs: net 360° after an additional 180°.
- Joint limits, friction, solver settings, and body lock are unchanged.
- Phi is a bounded torque-admittance command; theta remains held position.
- The base gets 1/11 of total bend; each later A/B joint gets 1/22.
- Torque is limited to ±1.0 N·m and slewed at 10.0 N·m/s.
- Immense-stiffness gains: roll 20M/42k; yaw 30M/51k.
- Rigid-body damping is 5.0 linear / 25.0 angular; neutral theta is 0°.
- The downward-bending standing posture uses positive torque on all four legs.
- Disc-orientation PI feedback compensates load: Kp 1.2, Ki 4.0/s.

Validation boundary

The 1 N·m → 180° law is a simulated command mapping, not raw or calibrated physical torque. Leg-angle tracking is simulation-only; hardware validation, broad contact stability, and locomotion remain future tests.

Next Research Steps

01	**Physical gauge selection**	Choose a commercial gauge, substrate, adhesive, and temperature compensation strategy.
02	**Bench calibration**	Measure real strain and bridge voltage against known curvature and loading.
03	**Independent shape validation**	Compare reconstructed curvature with motion capture or vision—not the same pose source.
04	**Loaded and directional trials**	Repeat across θ, negative torque, payloads, disturbances, and repeated cycles.
05	**Uncertainty and statistics**	Report repeatability, hysteresis, drift, confidence intervals, and failure limits.

Current Outcome

A reproducible six-disc research platform now connects

CONTROL	Position + torque admittance + minimum-energy circle
SHAPE	Synchronized five-interval constant-curvature target
SENSING	Six pose-derived strain channels + bridge voltages
INTEGRATION	Verified limb reused in a five-member quadruped
EVIDENCE	Automated tests and research documentation

Active boundary: ±1.0 N·m and ±180° • Next milestone: physical sensor calibration